Program overview

14-Dec-2022 11:11

Year	**2022/2023**
Organization	**Architecture and the Built Environment**
Education	**Minors BK**

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Minors BK 2022			
BK-MI-141-22 Stageminor			
BK5STA	Work in Practice: Architecture	20	
BK5STB	Work in Practice: Building Technology	20	
BK5STPO	Internship: Personal development	5	
BK5STR	Work in Practice: Management in the Built Environment	20	
BK5STU	Work in Practice: Urbanism	20	
BK5STVA	Report Work in Practice: Architecture	5	
BK5STVB	Report Work in Practice: Building Technology	5	
BK5STVR	Report Work in Practice, Management in the Built Environment	5	
BK5STVU	Report Work in Practice: Urbanism	5	
BK-MI-193-22	**BK-MI-193-22 Cities Migration and Socio-Spatial Inequality (CMSI)**		
BK7470	CMSI Lecture Series and Review Paper	6	
BK7471	CMSI Collaborative Project: Tackling Spatial Inequality	6	
BK7472	CMSI Engaging with Practice of Spatial Inequality	3	
BK-MI-197-22	**BK-MI-197-22 Spatial Computing in Architectural Design**		
BK7083	Computational Design Studio	9	
BK7084	Computational Simulations	6	
BK-MI-198-22	**BK-MI-198-22 Architecture Presentation: Visions Reviewed**		
BK7140	Presenting Project Facts	3	
BK7141	Presenting Project Visions	3	
BK7142	Presenting Project Prospects	9	
BK-MI-203-22	**BK-MI-203-22 Archineering Q1**		
BK7460-13	Archineering 1	15	
BK-MI-204-22	**BK-MI-204-22 Archineering Q2**		
BK7461	Archineering 2	15	
BK-MI-209-22	**BK-MI-209-22 Spaces of Display Q1**		
BK7064	Spaces of Display 1	10	
BK7066	Tools for Spaces of Display 1	5	
BK-MI-210-22	**BK-MI-210-22 Spaces of Display Q2**		
BK7065	Spaces of Display 2	10	
BK7067	Tools for Spaces of Display 2	5	
BK-MI-211-22	**BK-MI-211-22 Heritage and Design Q1**		
BK7550	Landscape and Transition	5	
BK7551	History of Dutch Cities and Landscapes	5	
BK7555	City and Transformation	5	
BK-MI-212-22	**BK-MI-212-22 Heritage and Design Q2**		
BK7552	Heritage: Theory and Practice	5	
BK7553	Architecture and Re-use	5	
BK7554	History of Dutch architecture and art	5	
BK-MI-2016-22	**BK-MI-216-22 Sustainable Urbanism: The Green-Blue City**		
BK7215	Basic techniques for sustainable urban design	5	
BK7225	City and Public Space: Sustainable Urban Design	10	
BK-MI-231-22 (Re)Imagining Port Cities: Understanding Space, Society and Culture			
BK7940	(Re)imagining Port Cities: Understanding Space, Society and Culture Q1	15	
BK7941	(Re)imagining Port Cities: Understanding Space, Society and Culture Q2	15	
BK-MI-235-22 Spatial Computing for Digital Twinning			
BK7944	Spatial Computing for Digital Twinning	15	
BK-MI-236-22 Timeless Typologies - "Remodelling Architecture"			
BK7942	Culture & Understanding	5	
BK7943	Design & Remodelling	10	

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Minors BK 2022

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Education	**Minors BK**

BK-MI-141-22 Stageminor

BK5STA	Work in Practice: Architecture	20
Course Coordinator	Dr. O. Caso	
Responsible for assignments	Dr. O. Caso	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	

BK5STB	Work in Practice: Building Technology	20
Course Coordinator	Ir. F.R. Schnater	
Responsible for assignments	Ir. F.R. Schnater	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	

BK5STPO	Internship: Personal development	5
Course Coordinator	Dr. J.L. Heintz	
Instructor	Prof.dr. P.W.C. Chan	
Responsible for assignments	Dr. J.L. Heintz	
Education Period	1 3 Different, to be announced	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	

BK5STR	Work in Practice: Management in the Built Environment	20
Course Coordinator	K.B.J. Van den Berghe	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	
Special Information	The maximum marking period is 10 work days.	

BK5STU	Work in Practice: Urbanism	20
Course Coordinator	R.J. van der Veen	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	

BK5STVA	Report Work in Practice: Architecture	5
Course Coordinator	Dr. O. Caso	
Responsible for assignments	Dr. O. Caso	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	

BK5STVB	Report Work in Practice: Building Technology	5
Course Coordinator	Ir. F.R. Schnater	
Responsible for assignments	Ir. F.R. Schnater	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	

BK5STVR	Report Work in Practice, Management in the Built Environment	5
Course Coordinator	K.B.J. Van den Berghe	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	

BK5STVU	Report Work in Practice: Urbanism	5
Course Coordinator	R.J. van der Veen	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	

Year	**2022/2023**
Organization	**Architecture and the Built Environment**
Education	**Minors BK**

BK-MI-193-22

BK7470	CMSI Lecture Series and Review Paper	6
Course Coordinator	Dr. R.J. Kleinhans	
Instructor	Prof.dr. M. van Ham	
Instructor	A. Petrovi	
Instructor	Dr. C. Cottineau	
Instructor	Dr. T. Verbeek	
Responsible for assignments	Dr. R.J. Kleinhans	
Contact Hours / Week x/x/x/x	6 hours per week, starting from week 1.1 and ending in week 1.9.	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	This first course of the minor Cities, Migration and Socio-Spatial Inequality (CMSI) offers interactive lectures introducing the context, themes and key concepts in the minor: socio-spatial inequality, spatial justice, migration, diversity, and resilience. For the interactive lectures, background literature will be provided on Brightspace. In turn, students will prepare statements and questions in relation to each lecture. Some of the lectures will be provided by instructors from other universities: Prof. dr. P.W.A. Scholten (Erasmus University Rotterdam - Public Administration) and Prof. dr. M. Schrover (Leiden University - Migration History). After a few weeks, students will select a theme or concept based on the lectures and write a literature review paper (max. 4200 words) with a research question that is approved by the instructors. This review should not only include a discussion of one of the key concepts, but also an analysis of its spatial implications. During the paper writing process, regular feedback will be provided by instructors. Each student will be assigned to a paper tutor for individual feedback (with a maximum of five students per instructor in order to enable rapid responses).	
Course Contents Continuation	This course is part of the 15 ECTS CMSI minor intended for students who are highly motivated to develop an interdisciplinary perspective on socio-spatial inequality, spatial justice, migration, diversity and resilience. Students from different backgrounds will acquire a thorough understanding of these concepts and learn how to analyse related questions from the academic disciplines of urban geography, sociology, urban planning and design, and public administration. Students will also be trained to apply this knowledge in plans with socio-spatial strategies for intervention in and redesign of concrete planning cases. CMSI Minor participants must have a passion for social scientific research and combine a strong academic curiosity with a determination to apply interdisciplinary knowledge in real-life situations of complex urban planning cases in the Netherlands.	
Study Goals	After this course, the students are able to: - Define the key concepts socio-spatial inequality, spatial justice, resilience, diversity, and identity; - Discuss the meaning of these concepts in the context of their theoretical background and the broader societal context; - Identify how socio-spatial inequality, migration and diversity manifest themselves on the level of cities and neighbourhoods, and provide examples of this manifestation; - Synthesize the relevant literature for a chosen theme or concept and analyse its spatial implications.	
Education Method	Interactive lectures, prepared pitches and discussions, role plays, literature study and writing a review paper. After each lecture, students can also individually submit a one-minute paper to the course instructor (by email), outlining their main lessons learned and identifying remaining questions. Students will also deliver (in pairs) short pitches, called 'Your Take on Social Inequality'. In these pitches, students choose and explain specific news items or topics related to the overall minor theme, and spark debate in class.	
Literature and Study Materials	Background literature will be provided on Brightspace. Students are required to buy the following book (approx. 25 euros): Hajer, M. A., Pelzer, P., Van Den Hurk, M., Ten Dam, C., & Buitelaar, E. (2020). Neighbourhoods for the future: a plea for a social and ecological urbanism. Trancity Valiz.	
Assessment	Instructor feedback on the literature review paper will be provided in writing and face-to-face at scheduled moments during the writing process. The summative assessment is the actual grading of the paper. Students should achieve a minimum grade of 5.8 for the paper before the final grade for the minor will be determined.	
Tags	Building & Spatial Development Policy Analysis Sustainability	
Period of Education	Quarter 1	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	-	

BK7471	CMSI Collaborative Project: Tackling Spatial Inequality	6
Course Coordinator	Dr. R.J. Kleinhans	
Instructor	Dr.ir. M.J. van Dorst	
Instructor	Dr. R.J. Kleinhans	
Instructor	Dr. T. Verbeek	
Responsible for assignments	Dr. R.J. Kleinhans	
Contact Hours / Week x/x/x/x	5 hours per week, starting from week 1.2 and ending in week 1.10.	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	This second course of the minor 'Cities, Migration and Socio-Spatial Inequality' consists of interdisciplinary group work, in which students from different backgrounds work together on a real-life case study with a complex set of problems related to inequality, diversity and migration, in an urban setting in Rotterdam. In fact, the case study will be embedded in the Urban Living Lab that is currently operational in Rotterdam (see also BK7472). The case study will include both action research problem analysis and stakeholder identification (partly within BK7472) and planning and design, i.e. creating a sociospatial strategy that addresses the identified problems and opportunities. Obviously, students are encouraged to align their literature review papers (BK7470) with the research part of the project. The group work extends over most of the length of the minor and starts with forming a project team of approximately three students. Each team will draft a project plan, outlining the chosen topic, research and design approach. Towards the end of the minor, the teams will present their case study analysis and intervention plan, process the feedback from instructors and fellow students, and subsequently submit a final project report.	
Course Contents Continuation	This course is part of the 15 ECTS CMSI minor intended for students who are highly motivated to develop an interdisciplinary perspective on socio-spatial inequality, spatial justice, migration, diversity and resilience. Students from different backgrounds will acquire a thorough understanding of these concepts and learn how to analyse related questions from the academic disciplines of urban geography, sociology, urban planning and design, and public administration. Students will also be trained to apply this knowledge in plans with socio-spatial strategies for intervention in and redesign of concrete planning cases. CMSI Minor participants must have a passion for social scientific research and combine a strong academic curiosity with a determination to apply interdisciplinary knowledge in real-life situations of complex urban planning cases in the Netherlands.	
Study Goals	After this course, the students are able to: - Analyse the nature of socio-spatial inequality, diversity, spatial justice, migration, and resilience and their effects on various spatial (mostly local) levels and from an interdisciplinary perspective; - Assess the planning, governance and design implications of socio-spatial inequality, migration and diversity at the urban and neighbourhood level; - Develop plans with socio-spatial strategies for intervention.	
Education Method	Students will be assigned to interdisciplinary project teams of three students by the course co-ordinator. Each team will draft a project plan, outlining the chosen topic, research and design approach. Instructors who will provide coaching and feedback to the students in consecutive meetings. Towards the end of the minor, the teams will present their case study analysis and intervention plan, process the feedback from instructors and fellow students, and subsequently submit a final project report.	
Assessment	Instructor feedback on the research and design process and outcomes will be provided at scheduled moments during the project. The summative assessment is the actual grading of the project report. Students should achieve a minimum grade of 5.8 for the project report before the final grade for the minor will be determined.	
Tags	Building & Spatial Development Fieldwork Policy Analysis Sustainability	
Period of Education	Quarter 1	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	-	

BK7472	CMSI Engaging with Practice of Spatial Inequality	3
Course Coordinator	Dr. R.J. Kleinhans	
Instructor	Dr. R.J. Kleinhans	
Instructor	Dr. T. Verbeek	
Responsible for assignments	Dr. R.J. Kleinhans	
Contact Hours / Week x/x/x/x	4 hours per week, starting from week 1.1 and ending in week 1.5.	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	This third course in the minor Cities, Migration and Socio-Spatial Inequality (CMSI) consists of an action research approach in a neighbourhood on the South Bank of Rotterdam. Using a range of methods, students will collect data that enables an integrated, multidisciplinary analysis of this neighbourhood, focussing on socio-spatial inequality, diversity and social resilience (in Dutch: veerkracht). The multidisciplinary analysis will uncover socio-spatial inequality and diversity across several domains, ranging from housing, public space to education, employment and participation and safety. This analysis will form the basis of the strategic plan that will be created in the BK7471 course. The data collection for BK7472 will be embedded in one of the Urban Living Labs currently operational on the South Bank of Rotterdam. As such, students will also engage with practitioners, including policymakers of the local authorities, staff members of housing associations, welfare workers or representatives of civil society organisations. As part of this engagement with local professionals, students will visualise the collected data. The collected data and analysis are the main inputs for the collaborative strategic plans (BK7471).	
Course Contents Continuation	This course is part of the 15 ECTS CMSI minor intended for students who are highly motivated to develop an interdisciplinary perspective on socio-spatial inequality, spatial justice, migration, diversity, and resilience. Students from different backgrounds will acquire a thorough understanding of these concepts and learn how to analyse related questions from the academic disciplines of urban geography, sociology, urban planning and design, and public administration. Students will also be trained to apply this knowledge in plans with socio-spatial strategies for intervention in and redesign of concrete planning cases. Minor participants must have a passion for social scientific research and combine a strong academic curiosity with a determination to apply interdisciplinary knowledge in real-life situations of complex urban planning cases in the Netherlands.	
Study Goals	After this course, students are able to: 1. Create an integrated and multidisciplinary spatial analysis of social inequality in a diverse urban neighbourhood; 2. Visualise and discuss their collected data and analyses with local practitioners; 3. Understand how urban practitioners address (local) issues related to spatial inequality, diversity, migration, social resilience, and identity.	
Education Method	The course provides instruction and training in action research methods, in particular observations, street interviews, in-depth interviews and desk research. There will also be attention to visualization. The training offers students a kick-start for their research and data collection for their strategic plan (BK7471). Throughout the course, supervision and feedback will be provided by the course instructors.	
Assessment	Throughout the course, supervision and feedback will be provided by the course instructors. The summative assessment is the actual grading of the multidisciplinary analysis report. Students should achieve a minimum grade of 5.8 for this report before the final grade for the minor will be determined.	
Tags	Building & Spatial Development Fieldwork Policy Analysis Sustainability	
Period of Education	Quarter 1	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	-	

Year	**2022/2023**
Organization	**Architecture and the Built Environment**
Education	**Minors BK**

BK-MI-197-22

BK7083	Computational Design Studio	9
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	Prof.dr.ir. I.S. Sariyildiz	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	This course is a Generative Design studio focused on Gamified Mass-Customization of Housing. It is offered as a part of the Minor Spatial Computing in Architectural Design [BK-MI-197], which is the second half of the Minor Spatial Computing for Sustainable Development. For more information about the complete minor degree programme check the web-page: https://www.tudelft.nl/bk/studeren/minoren-en-keuzevakken/spatial-computing-for-sustainable-development. The assignment of this course is to computationally design a housing complex, given a set of user requirements, 3D geospatial data of the site, and spatial/environmental quality criteria. In this course, the fundamentals of computational design are taught. Mathematical subjects from Computer Geometry, Linear Algebra, Topology, and Graph Theory, as well as the fundamentals of Artificial Intelligence for architectural design will be reviewed or covered. Students will learn how to: set up a systematic computational design process; design basic algorithms for procedural modelling of spatial designs; and implement the algorithms by means of [visual] programming in Python (NumPy), only by using open-source digital technologies. To get an impression of some of the results achieved in the past years visit this page: https://genesis-lab.dev/courses/spatialcomputing/.	
Study Goals	Having followed this course, the student is expected to be able to set up a systematic computational design process for meeting the design requirements and achieving high levels of performance as to a set of quality criteria/objectives.	
Education Method	research-based learning in design studios, mathematical lectures combined with hands-on programming workshops, and consultations in study groups.	
Assessment	The assessment is based on the quality of deliverables: a computational design process and the schematic design of a housing complex generated computationally. Assessment Types: Design Examination, Analytical Assignment, and Writing Assignment	
Tags	Algorithmics Artificial intelligence Building & Spatial Development Lineair Algebra Mathematics Programming	
Period of Education	Q2	

BK7084	Computational Simulations	6
Course Coordinator	Prof.dr. E. Eisemann	
Course Coordinator	Dr. R. Marroquim	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic mathematical skills (vector calculus, linear algebra).	
Course Contents	This course investigates the use of simulation and optimization algorithms, with a focus on image synthesis and light transport, which can be of use in the context of interior and architectural design, as well as urban planning.	
Study Goals	The students will be able to apply aspects of physically-based light simulation, transformations and procedural content generation to architectural design and urban planning. They will learn how to code simple prototypes themselves to verify their knowledge.	
Education Method	The course is organized in lectures and practicals. Further, there are exercise and instruction sessions. The content of the lectures will be partly reproduced in the practicals and a final project will integrate different elements of the practicals.	
Computer Use	The course will contain a programming component. Coding skills can be acquired during the course, which will include introductory lectures.	
Assessment	The course is evaluated by two means: an exam (can be multiple choice) and a final project (in small groups). The exam accounts for 35% of the final grade, and the final project accounts for 65%, The exam resit might take form of an oral exam.	
Period of Education	Quarter 2	

Year	**2022/2023**
Organization	**Architecture and the Built Environment**
Education	**Minors BK**

BK-MI-198-22

BK7140	Presenting Project Facts	3
Course Coordinator	P.A. Koorstra	
Course Coordinator	M.G. Vink	
Instructor	W.C. Yung	
Instructor	M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	12 hours a week for two weeks	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	This course is part of the half minor Architecture Presentation Visions Reviewed, BK-MI-198. Students will choose a group of projects from a number of given architectural precedents. In small groups (3-4), they will work towards a high level of project understanding and present their insights and interpretations by means of oral presentations, posters, analytical drawings and/or collages. Develop project understanding, its' design narrative, contextual position and representations. Develop analysis, presentation and representation skills: slide design analytical drawings, info graphs and posters, project discourse reporting own progress (blog) oral presentation.	
Study Goals	After finishing this course, the student has: acquired knowledge, insights and skills in the domains of project analysis and presentation, developed skills in the effective representation and communication of a chosen case project, using various representation techniques, the ability to work individually and in a team, carrying out analyses, documenting results, visualizing proposals and bringing these across to a professional audience.	
Education Method	Course themes will be introduced in the syllabus and lectures. Consults will be held in small groups and individually - depending on the assignment. There will be two contact moments per week. We will make use of BrightSpace and personal online blogs to submit share and comment on results. Assignments are executed in a design studio setting. On Tuesdays and Thursdays students are expected to be in the studio. For the other days it is recommended to work in the studio, but it is up to the student to decide what works best. Students will keep a personal blog/site to present own results, group results and reflections on their progress. The blog needs to be updated at the end of each week.	
Assessment	The minor consists of a series of integrated exercises which are all assessed separately. All these assignments - except for the online blog - are assessed through oral presentation guided by the requested presentation products. These are consequently: - Pecha Kucha Slide show - A2 Poster - Written Storyline & Visual Storyboard + Planning - Vision Presentation with a product developed through a chosen skill (drawing, painting, model, render, photography, video, text) - Exhibition + Exhibition product and booklet - Blog with weekly reflection on the process - textual and visual. The tutors make the assessment based on: - completeness of requested products - readability of the representation products - quality of the representation products - creativity and coherence in the representation products - relevance of the produced products in relation to the studied project - coherence between oral presentation and representation products - clarity of oral presentation The marks for the various exercises are translated to the grades per course as follows: - BK7140 - 3 ECTS : Pecha Kucha + Poster (50%, 50%) - BK7141 - 3 ECTS : Exhibition + Blog (50%, 50%) - BK7142 - 9 ECTS : Storyline + Vision Presentation (25%, 75%)	
Special Information	Coordinator	
Period of Education	Q2 of the first semester.	
Leerstoel	Form & Modelling Studies, Arcihtecture	
Minimum number of participants	12	
Maximum number of participants	30	

BK7141	Presenting Project Visions	3
Course Coordinator	P.A. Koorstra	
Course Coordinator	M.G. Vink	
Instructor	W.C. Yung	
Instructor	M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	12 hours a week for two weeks	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor 1-4	
Course Contents	This course is part of the half minor Architecture Presentation Visions Reviewed, BK-MI-198. Students will choose a project from a number of given inspirational architecture visions (e.g. New Babylon by Constant) which will be interpreted in new ways and represented through various media. This part of the minor, concerns the reflection on this process. This will be done along the way; by keeping up an online blog (individual). This is done through text and images and requires the skill of writing, documenting and reflection. The last part of the course, forms a reflection on the results as a group (3-4, organized per vision). The members of each Pecha Kucha-group at the start of the minor, will bring their individual works together in the curation of an exhibition. Students will thus develop reflective presentation skills through: individual writing on the personal lessons and findings throughout the process documenting of the process - images, drawings, models, thoughts. online blog design group group discussion and brainstorming on communal narrative exhibition design and building design of explanatory exhibition booklet oral presentation	
Study Goals	After finishing this course, the student has: developed the ability to document, explain and reflect on his/her own learning process and produced work both in text and in image in a clear and inspiring way. the ability to synthesize the work of individuals into a communal narrative (bring individual work together in a groupwork. the ability to translate and curate a variety of representation products by multiple individuals and through different representational means through the ideation and production of an exhibition.	
Education Method	Course themes will be introduced in the syllabus and lectures. Consults will be held in small groups and individually - depending on the assignment. There will be two contact moments per week. We will make use of BrightSpace and personal online blogs to submit share and comment on results. Assignments are executed in a design studio setting. On Tuesdays and Thursdays students are expected to be in the studio. For the other days it is recommended to work in the studio, but it is up to the student to decide what works best. Students will keep a personal blog/site to present own results, group results and reflections on their progress. The blog needs to be updated at the end of each week.	
Assessment	The minor consists of a series of integrated exercises which are all assessed separately. All these assignments - except for the online blog - are assessed through oral presentation guided by the requested presentation products. These are consequently: - Pecha Kucha Slide show - A2 Poster - Written Storyline & Visual Storyboard + Planning - Vision Presentation with a product developed through a chosen skill (drawing, painting, model, render, photography, video, text) - Exhibition + Exhibition product and booklet - Blog with weekly reflection on the process - textual and visual. The tutors make the assessment based on: - completeness of requested products - readability of the representation products - quality of the representation products - creativity and coherence in the representation products - relevance of the produced products in relation to the studied project - coherence between oral presentation and representation products - clarity of oral presentation The marks for the various exercises are translated to the grades per course as follows: - BK7140 - 3 ECTS : Pecha Kucha + Poster (50%, 50%) - BK7141 - 3 ECTS : Exhibition + Blog (50%, 50%) - BK7142 - 9 ECTS : Storyline + Vision Presentation (25%, 75%)	
Period of Education	Q2	
Leerstoel	Form & Modelling Studies, Architecture	
Minimum number of participants	12	
Maximum number of participants	30	

BK7142	Presenting Project Prospects	9
Course Coordinator	P.A. Koorstra	
Course Coordinator	M.G. Vink	
Instructor	W.C. Yung	
Instructor	M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	12 hours a week for two weeks	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor 1-4	
Course Contents	This course is part of the half minor Architecture Presentation Visions Reviewed, BK-MI-198. Students will design a small scale building, related to the design themes of the given precedents in the first exercises. This is first done by conceptual modelling, handsketching and writing. For this specific part of the Minor, students will create an evocative moving image - a short film (max 6 min) which communicates the character of their design idea. The aim is to develop a storyline, storyboard and find an appropriate presentation styles through chosen artistic means (animation, handdrawings, modelling, filming, painting, ...). Each student will present their produced work in an oral presentation, guided by a digital slideshow. This is all done on an individual basis.	
Study Goals	After finishing this course, the student has: developed new insights and presentations concerning the further expression and re-interpretation of the case study project, developed creative and technical skills in the effective representation and communication of a chosen case project, using various representation techniques. the ability to work individually and critically, carrying out analyses, documenting results, visualizing proposals and bringing these across to a professional audience. Academic competences Synthesis of project facts, character, ideas and ideals. Transformation of the case project to a contemporary or futuristic project presentation.	
Education Method	Course themes will be introduced in the syllabus and lectures. Consults will be held in small groups and individually - depending on the assignment. There will be two contact moments per week. We will make use of BrightSpace and personal online blogs to submit share and comment on results. Assignments are executed in a design studio setting. On Tuesdays and Thursdays students are expected to be in the studio. For the other days it is recommended to work in the studio, but it is up to the student to decide what works best. Students will keep a personal blog/site to present own results, group results and reflections on their progress. The blog needs to be updated at the end of each week.	
Assessment	The minor consists of a series of integrated exercises which are all assessed separately. All these assignments - except for the online blog - are assessed through oral presentation guided by the requested presentation products. These are consequently: - Pecha Kucha Slide show - A2 Poster - Written Storyline & Visual Storyboard + Planning - Vision Presentation with a product developed through a chosen skill (drawing, painting, model, render, photography, video, text) - Exhibition + Exhibition product and booklet - Blog with weekly reflection on the process - textual and visual. The tutors make the assessment based on: - completeness of requested products - readability of the representation products - quality of the representation products - creativity and coherence in the representation products - relevance of the produced products in relation to the studied project - coherence between oral presentation and representation products - clarity of oral presentation The marks for the various exercises are translated to the grades per course as follows: - BK7140 - 3 ECTS : Pecha Kucha + Poster (50%, 50%) - BK7141 - 3 ECTS : Exhibition + Blog (50%, 50%) - BK7142 - 9 ECTS : Storyline + Vision Presentation (25%, 75%)	
Period of Education	Q2	
Leerstoel	Form & Modelling Studies, Architecture	
Minimum number of participants	12	
Maximum number of participants	30	

Year	**2022/2023**
Organization	**Architecture and the Built Environment**
Education	**Minors BK**

BK-MI-203-22

BK7460-13	Archineering 1	15
Course Coordinator	Ir. R.R.J. van de Pas	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Parts	Design Process, Experimenting, Digital Fabrication, Materialization & Detailing	
Course Contents	The minor Archineering focuses on materialisation as an essential part of any design product. In a number of intense and very hands-on prototyping exercises, more insight in the personal design process and integration of technology and design will be acquired in this minor. The more experienced designer students will be able to improve themselves on specific aspects like Digital Fabrication and 3D printing. But also students new in the field of architecture will explore and discover the fun of designing, sometimes even scale 1:1!	
Study Goals	In the minor Archineering you will learn to Foster design results, in which the relation between design and technology is a key theme; Formulate, use and develop a guiding theme to strengthen the coherence within your design project on all scale levels; Translate a design task in a complete and technically engineered draft proposal within 2 weeks; Do research on the technology aspects: construction, climate design, digital manufacturing and detailing and be able to experiment with the design tools you find in your own design project; Analyse buildings and compare them on various generic architectural aspects, to translate this in diagrams and to apply these as a design tools in your own design process; Build, test and develop several prototypes and models on different information levels; Be able to make explicit the generic elements of the design process and use these to reflect on your own design process;	
Education Method	Design Projects: In Archineering 1 students are gradually introduced to the generic elements of the design process, by means of several seminars and a series of small, two or three week design exercises often with a given topic and guiding theme. The integration of materialisation and architecture is examined by focusing on one single aspect in every design exercise, for example: construction or climate design. Excursions and Workshops: Each design exercise starts with an excursion, lecture or workshop on the particular technical aspect that is focussed on (construction/environment design/detailing). These make the students playfully familiar with the specific technical challenges within the work field of the central theme. After this, the student will practice and experiment with the knowledge gained in his own design project. Reflection: After the design assignments a reflection report is written and handed in.	
Prerequisites	This course is open to any Bachelor student who wants to delve more into the materialization and engineering of their design plans, improve their design skills and get more insight in the design process. Students from every faculty at TU Delft should have received the adequate knowledge prior to entering this minor. Also, students from other universities or in HBO education can apply for the minor. In this case a small portfolio and CV is asked, to check if design skills and frame of reference are sufficient. This minor consists of 2 quarters (both of 15 ECs); each with separate content. For students of the Faculty of Architecture and the Built Environment TU Delft it is possible to follow either just quarter 1 or just quarter 2, or to follow both quarter 1 and 2. For all other students who are admissible to this minor, it is only possible to follow either quarter 1, or both quarter 1 and 2. Please be aware that each quarter has a separate minor code. If you would like to follow both quarter 1 and 2, you should register for both minor codes in Osiris.	
Assessment	Design proposal * = Verbal presentation with poster or beamer and lots of sketch models!!! Reflection Paper ** = Analyses in text, photos and diagrams. * > We do quite a lot of design projects, even compared to other courses at The Faculty of Architecture and the Built Environment. This is to let the students really experience as much as possible the different aspects of the design process. By training designing, engineering and prototyping in a number of short exercises and explicitly studying and reflecting on the design process, more insight and understanding (in the personal design process and integration of technology & design) will be acquired. ** > students do an overall reflection at the end of Archineering 1.	
Period of Education	Quarter	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations see: http://kwaliteitszorg.bk.tudelft.nl/	

Year	**2022/2023**
Organization	**Architecture and the Built Environment**
Education	**Minors BK**

BK-MI-204-22

BK7461	Archineering 2	15
Course Coordinator	Ir. R.R.J. van de Pas	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Parts	Design Process, Research, Digital Fabrication, Materialization & Detailing	
Course Contents	The minor Archineering focuses on materialisation as an essential part of any design product. In a number of intense and very hands-on prototyping exercises, more insight in the personal design process and integration of technology and design will be acquired in this minor. The more experienced designer students will be able to improve themselves on specific aspects like Digital Fabrication and 3D printing. But also students new in the field of architecture will explore and discover the fun of designing, sometimes even scale 1:1!	
Study Goals	In this minor we will work on one research & design task and you will learn to: foster design results, in which the relation between design and materialisation is a key theme; translate a design task in a complete and technically engineered proposal; do research on the technology aspects: construction, climate design, digital manufacturing and detailing, and be able to experiment with these in your own design project; build, test and develop several prototypes and models on different information levels and scale levels;	
Education Method	Design Project: In Archineering 2 the knowledge and design methodology gained in Archineering 1 is taken into to practice more independently by working on one integrated design project with greater complexity (construction and climate design and materialization & detailing). Reflection: After the design assignment a reflection report is written and handed in.	
Prerequisites	The prior knowledge from Archineering 1 is essential for taking part in Archineering 2. This minor consists of 2 quarters (both of 15 ECs); each with separate content. For students of the Faculty of Architecture and the Built Environment TU Delft it is possible to follow either just quarter 1 or just quarter 2, or to follow both quarter 1 and 2. For all other students who are admissible to this minor, it is only possible to follow either quarter 1, or both quarter 1 and 2. Please be aware that each quarter has a separate minor code. If you would like to follow both quarter 1 and 2, you should register for both minor codes in Osiris.	
Assessment	Design proposal = Verbal presentation with poster or beamer and lots of sketch models!!! Reflection Paper = Analyses in text, photos and diagrams.	
Period of Education	Quarter	
Leerstoel	The minor Archineering is a coherent package of courses in the field of Architectural Engineering (aE). It is strongly related to the aE-MSc 1 course Buckey Lab, the aE-MSc 2 course van Gezel tot Meester and the aE- MSc 3+4 Graduation Studio Intecture.	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations see: http://kwaliteitszorg.bk.tudelft.nl/	

Year	**2022/2023**
Organization	**Architecture and the Built Environment**
Education	**Minors BK**

BK-MI-209-22

BK7064	Spaces of Display 1	10
Course Coordinator	Ir. L.M.M. de Wit	
Instructor	Ir. S. Pietsch	
Responsible for assignments	Ir. L.M.M. de Wit	
Contact Hours / Week x/x/x/x	X/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Required for	Bachelor students Architecture and Industrial Design (TU Delft and Eindhoven) Extern HBO: Design Acadamy Eindhoven Art academies; graduates in Architectural Design, interior architecture	
Expected prior knowledge	Bachelor 1 to 4 or equivalent. or: Three years of HBO or advanced Bachelor in abovementioned educations. Recommended skills: Revit, Rhino, Autocad, Illustrator, Sketchup or equivalent.	
Summary	The Chair of Interiors Buildings Cities is concerned with making buildings, in places, for people. It conceives of the city, at each scale, as a work of architecture and, hence, the responsibility of the architect. This developing discussion of the chair is contextualized through an annual theme. During this interiors minor, students design a (small) exhibition space (Q1) and a fully detailed fragment of the building or the interior (Q2). We emphasize on the use of scale models as a means of research. This minor consists of 2 quarters (both of 15 ECs); each with separate content. For students of the Faculty of Architecture and the Built Environment TU Delft it is possible to follow either just quarter 1 or just quarter 2, or to follow both quarter 1 and 2. For all other students who are admissible to this minor, it is only possible to follow either quarter 1, or both quarter 1 and 2. Please be aware that each quarter has a separate minor code. If you would like to follow both quarter 1 and 2, you should register for both minor codes in Osiris.	
Course Contents	Q1 BK-MI-209 During the first quarter, students design a small exhibition space for a carefully selected artefact and its users at a specific location in a city (BK7064 Spaces of Display 1). The Tools Course (BK7066 Tools for Spaces of Display 1) focuses on specific skills that contribute to a coherent design project. Students learn how to work within the complex combination of these 'tools'. Various workshops and lectures focus on the position of the designer, the relationship with the city, the design of a space within the field of the exhibition branch, and the analysis of reference projects based on relevant themes. Based on a thorough analysis of the location in the city, the choice is made for the artefact/product and a specific user group. This allows the possibility to explore the full range of related interventions. Q2 BK-MI-210 During the second quarter, students design and build a fragment of the building on different scale levels (1:10/1:5 and sometimes even 1:1) based on the outcome of their space or display (or that of a fellow student) designed in Q1. This design project (BK7065 Spaces of Display 2) runs parallel to a Tools course (BK7067 Tools for Spaces of Display 2). The workshops and lectures in this quarter make a substantial contribution to the theme of the project.	
Study Goals	At the end of Q1 you will: have acquired knowledge, insights and skills in the domains of architectural and interior design, analysis, and presentation; have the skill to establish the connection between idea, function and material elaboration of a small scale design; have knowledge of different cultural and social conditions, and awareness of their implication and effects on architecture and the interior; be able to develop a consistent architectonic / interior design. This covers the materialization of the overall space as well as the detailing have developed abilities in the targeted analysis of design precedents and the effective representation and communication of findings, using various representation techniques as well as digital and physical models.	
Education Method	Excursion Design Tutorials Lectures Workshops	
Literature and Study Materials	Recommended literature: De Narratieve Ruimte ; Herman Kossmann Strategies of Display ; Julia Noordegraaf Exhibition Design - portfolio ; Philip Hughes Exhibition Design; Basic Interior Design 02 ; Pam Locker	
Assessment	Design project BK7064: final and midterm presentation with drawings/posters and models.	
Special Information	The excursion in the first week will be to a city in the Netherlands or Belgium. You will have to pay for your own travel expenses and entrance fees.	
Period of Education	Quarter 1	
Concept Schedule	Tuesday- and Fridaymornings designtutorials	
Leerstoel	Interiors Buildings Cities	
Minimum number of participants	18 students	
Maximum number of	27 students	

participants	
Course evaluation	For the course evaluations see: http://kwaliteitszorg.bk.tudelft.nl/

BK7066	**Tools for Spaces of Display 1**	**5**
Course Coordinator	Ir. L.M.M. de Wit	
Instructor	Ir. S. Pietsch	
Responsible for assignments	Ir. L.M.M. de Wit	
Contact Hours / Week x/x/x/x	X/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Required for	Bachelor students Architecture and Industrial Design (TU Delft and Eindhoven) Extern HBO: Design Acadamy Eindhoven Art academies; graduates in Architectural Design, interior architecture.	
Expected prior knowledge	Bachelor 1 to 4 or equivalent. or: Three years of HBO or advanced Bachelor in abovementioned educations. Recommended skills: Autocad, Illustrator, Sketchup or equivalent.	
Summary	The Chair of Interiors Buildings Cities is concerned with making buildings, in places, for people. It conceives of the city, at each scale, as a work of architecture and, hence, the responsibility of the architect. This developing discussion of the chair is contextualized through an annual theme. During this interiors minor, students design a (small) exhibition space with a fully detailed fragment of the building or interior. The minor consists of two quarters, each divided into two separate courses: The Design project and the Tools course	
Course Contents	The Tools Course (BK7066 Tools for Spaces of Display 1) focuses on specific skills that contribute to a coherent design project. Students learn how to work within the complex combination of these 'tools'. Various workshops and lectures focus on the position of the designer, the relationship with the city, the design of a space within the field of the exhibition branch, and the analysis of reference projects based on relevant themes. Based on a thorough analysis of the location in the city, the choice is made for the artifact/product and a specific user group. This allows the possibility to explore the full range of related interventions. During the workshops, specific knowledge of particular use to the design project will be gained from various discrete exercises. Experts will transfer their knowledge in a compact course in which students will be asked to make small studies in order to apply this knowledge directly. Important themes are location, the user and the artefact, quality of the perception of the interior and the materialization and detailing of space itself. We emphasize on the use of scale models as a means of research. This minor consists of 2 quarters (both of 15 ECs); each with separate content. For students of the Faculty of Architecture and the Built Environment TU Delft it is possible to follow either just quarter 1 or just quarter 2, or to follow both quarter 1 and 2. For all other students who are admissible to this minor, it is only possible to follow either quarter 1, or both quarter 1 and 2.	
Study Goals	- Has gained knowledge about the cultural and technical aspects of specific themes (for example: user research, materialization and detail) - Has learned to translate fresh knowledge into a design by doing a short practical study. - Is able to connect aspects like location, program, target group, functionality and temporariness to the atmospheric quality of building and space;	
Education Method	Lectures, Analysis, Intensive practical exercises with consults	
Assessment	Assignments, Lectures, Analysis and Workshops presented in a portfolio	
Period of Education	Quarter	
Leerstoel	Interiors Buildings Cities	
Minimum number of participants	18 students	
Maximum number of participants	27 students	
Course evaluation	For the course evaluations see: http://kwaliteitszorg.bk.tudelft.nl/	

BK-MI-210-22

BK7065	Spaces of Display 2	10
Course Coordinator	Ir. L.M.M. de Wit	
Instructor	Ir. S. Pietsch	
Responsible for assignments	Ir. L.M.M. de Wit	
Contact Hours / Week x/x/x/x	0/X/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor 1 to 4 or equivalent. or: Three years of HBO or advanced Bachelor in above mentioned educations. Recommended skills: Autocad, Illustrator, Sketchup or equivalent.	
Summary	The Chair of Interiors Buildings Cities is concerned with making buildings, in places, for people. It conceives of the city, at each scale, as a work of architecture and, hence, the responsibility of the architect. This developing discussion of the chair is contextualized through an annual theme. During this interiors minor, students design in Q1 a (small) exhibition space (BK7064) and in Q2 a fully detailed fragment of the building or the interior (BK7066). We emphasize on the use of scale models as means of research.	
Course Contents	During the second quarter, students design and build a fragment of the building or the interior on different scale levels (1:10/1:5 and sometimes even 1:1) based on the outcome of their space or display (or that of a fellow student) designed in Q1. This design project (BK7065 Spaces of Display 2) runs parallel to a Tool course (BK7067 Tools for Spaces of Display 2). The workshops and lectures in this quarter make a substantial contribution to the theme of the project. Q1 BK-MI-209 During the first quarter, students design a small exhibition space for a carefully selected artifact and its users at a specific location in a city (BK7064 Spaces of Display 1). The Tools Course (BK7066 Tools for Spaces of Display 1) focuses on specific skills that contribute to a coherent design project. Students learn how to work within the complex combination of these 'tools'. Various workshops and lectures focus on the position of the designer, the relationship with the city, the design of a space within the field of the exhibition branch, and the analysis of reference projects based on relevant themes. Based on a thorough analysis of the location in the city, the choice is made for the artifact/product and a specific user group. This allows the possibility to explore the full range of related interventions. Q2 BK-MI-210 During the second quarter, students design and build a fragment of the building or the interior on different scale levels (1:10/1:5 and sometimes even 1:1) based on the outcome of their space or display (or that of a fellow student) designed in Q1. This design project (BK7065 Spaces of Display 2) runs parallel to a Tools course (BK7067 Tools for Spaces of Display 2). The workshops and lectures in this quarter make a substantial contribution to the theme of the project. This minor consists of 2 quarters (both of 15 ECs); each with separate content. For students of the Faculty of Architecture and the Built Environment TU Delft it is possible to follow either just quarter 1 or just quarter 2, or to follow both quarter 1 and 2. For all other students who are admissible to this minor, it is only possible to follow either quarter 1, or both quarter 1 and 2. Please be aware that each quarter has a separate minor code. If you would like to follow both quarter 1 and 2, you should register for both minor codes in Osiris.	
Study Goals	At the end of Q2 you will: have the skill to establish the connection between idea, function and material elaboration of a fragment of the building / interior; have acquired knowledge, insights and skills in the domains of furniture / interior design, analysis and presentation; have developed abilities in the targeted analysis of design precedents and the effective representation and communication of findings, using various representation techniques as well as digital and physical models; be able to develop a consistent building/interior fragment. This covers the materialization of the object itself as well as the detailed scale; the processing of technical and aesthetic aspects plays an essential part in this. .can determine the quality of the perception of the interior by the use of artificial light.	
Education Method	design tutorials	
Literature and Study Materials	Recommended literature: De Narratieve Ruimte ; Herman Kossmann Strategies of Display ; Julia Noordegraaf Exhibition Design - portfolio ; Philip Hughes Exhibition Design; Basic Interior Design 02 ; Pam Locker	
Assessment	Design project BK7065: final and midterm presentation with drawings/posters and models.	
Period of Education	Quarter 2	
Concept Schedule	Design Tutorials on Tuesdays- and Friday mornings	
Leerstoel	Interiors Buildings Cities	
Minimum number of participants	18 students	
Maximum number of participants	27 students	
Course evaluation	For the course evaluations see: http://kwaliteitszorg.bk.tudelft.nl/	

BK7067	Tools for Spaces of Display 2	5
Course Coordinator	Ir. L.M.M. de Wit	
Instructor	Ir. S. Pietsch	
Responsible for assignments	Ir. L.M.M. de Wit	
Contact Hours / Week x/x/x/x	0/X/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	Bachelor students Architecture and Industrial Design (TU Delft and Eindhoven) Extern HBO: Design Acadamy Eindhoven Art academies; graduates in Architectural Design, interior architecture	
Expected prior knowledge	Bachelor 1 to 4 or equivalent. or: Three years of HBO or advanced Bachelor in abovementioned educations. Recommended skills: Autocad, Illustrator, Sketchup or equivalent.	
Summary	The Chair of Interiors Buildings Cities is concerned with making buildings, in places, for people. It conceives of the city, at each scale, as a work of architecture and, hence, the responsibility of the architect. This developing discussion of the chair is contextualized through an annual theme. During this interiors minor, students design a (small) exhibition space with a fully detailed fragment of the building or the interior. The minor consists of two quarters, each divided into two separate courses: The Design project and the Tools course	
Course Contents	The Tools Course (BK7067 Tools for Spaces of Display 2) focuses on specific skills that contribute to a coherent design project; a fragment of the building or the interior. Students learn how to work within the complex combination of these 'tools'. Various workshops and lectures focus on f.i.the analysis of reference projects based on relevant themes, the position of the designer and the joiner , the design of a fragment of the building or a display element. During the workshops, specific knowledge of particular use to the design project will be gained from various discrete exercises. Experts will transfer their knowledge in a compact course in which students will be asked to make small studies in order to apply this knowledge directly. Important themes are,the user of the space and the designed fragment, displaying products, the materialization and detailing of the fragment itself. We emphasize the use of scale models as a means of research. This minor consists of 2 quarters (both of 15 ECs); each with separate content. For students of the Faculty of Architecture and the Built Environment TU Delft it is possible to follow either just quarter 1 or just quarter 2, or to follow both quarter 1 and 2. For all other students who are admissible to this minor, it is only possible to follow either quarter 1, or both quarter 1 and 2.	
Study Goals	The student: - Has gained knowledge about the cultural and technical aspects of specific themes (for example: user research, lighting and detailing) - Has learned to translate fresh knowledge into a design by doing a short practical study. - Is able to determine the quality of the perception of the interior by using artificial light. - Is able to connect aspects like program, user group, functionality,temporariness and the connection of atmospheric quality of the fragment in the space.	
Education Method	Lectures, Analysis, Intensive practical exercises with consults	
Assessment	Assignments Lectures, Analysis and Workshops presented in a portfolio	
Period of Education	Quarter	
Concept Schedule	lectures, workshops on Tuesday afternoons	
Leerstoel	Interiors Buildings Cities	
Minimum number of participants	18 students	
Maximum number of participants	27 students	
Course evaluation	For the course evaluations see: http://kwaliteitszorg.bk.tudelft.nl/	

Year	**2022/2023**
Organization	**Architecture and the Built Environment**
Education	**Minors BK**

BK-MI-211-22

BK7550	Landscape and Transition	5
Course Coordinator	Dr.ir. G.A. Verschuure-Stuip	
Responsible for assignments	Dr.ir. G.A. Verschuure-Stuip	
Co-responsible for assignments	Dr. I. Nevzgodin	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Continuity and transformation are the keywords in our urban environment.The transformation of our built environment is of all ages and is currently one of the most urgent and challanging tasks in our society. In the design studio Landscape and Transition, you are asked to make a concept and design for a part of the green urban heritage in the city. The basis for an intervention is historic spatial analysis of the site, knowledge of the landscape architectural, urban and architectural history as well as current issues and problems in the urban public space. The challenge is to preserve monumental values and existing qualities as part of the sense of the place and combine this with new functions for up-to-date use. Studies of various methods of transformation of green heritage is part of the framework and will be studied and preserted in reference projects. During the course you will familiarise yourself with design skills, knowledge of city and architectural development of Dutch cities and landscapes, analysis methods and various theoretic approaches, as well as in cultural, societal and philosophical aspects. You will have the opportunity to gain knowledge in and to practise with aspects such as historical identity, value proposition and spatial intervention with objects and structures of cultural heritage in a historic environment.	
Study Goals	You will practise the interaction between history and design. After this course a students is able to successfully: - determine and draw the historic characteristics and current spatial qualities of specific cultural landscapes, green monumental heritage in word and drawings - Connect the characteristics of the historic urban landscape with new transformations and interventions. - make a simple landscape architectural plan for an intervention in a cultural landscape or urban green heritage site - reflect on the connection between preservation of the historic character, your plan and new interventions.	
Education Method	Design studio Landscape and Transition In the design studio, you are asked to produce a spatial design of a part of the site and work on specific details. To get to this design you will discuss and use various theories and methods in current and historic analysis as well as various design skills (architectural design, sketching, mood boards, collage, models, etc.).	
Assessment	The design studio will round up with a final presentation of the analysis and design as well as a presentation of reference projects.The course as well as the minor is successfully completed if you complete all study parts with grade 6 or higher.	
Special Information	The course Landscape and Transition is part of the minor City and Landscape: heritage and transformation in the first quarter. It is preliminary for the minor Design: Architecture and Heritage in the second quarter. The programme of the entire minor consists of one lecture series and two studio assignments, they account for a total of 15 study points whereby each part is worth 5 study points. BK7551 History of Dutch cities and Landscapes (lecture series) About the development of the Dutch city and the Dutch landscape. BK7555 Research Studio City and Transformation (studio assignment) Focuses on the analysis and assessment of the city and its buildings. The analysis must ultimately result in a cultural-historic value proposition. BK7550 Landscape and Transition (studio assignments) For the design studio Landscape and Intervention, a design must be made of the intervention of a public space, in a historic urban or landscape environment. The contact hours for the studio assignment is 8 hours per week. The contact hours for the lecture series can be maximum 8 hours per week. Included self-study the student is supposed to spend (15x28=) 420 hours on the whole course.	
Period of Education	Quarter 1	
Course evaluation	For the course evaluations see: http://kwaliteitszorg.bk.tudelft.nl/	

BK7551	History of Dutch Cities and Landscapes	5
Course Coordinator	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. R.J. Rutte	
Responsible for assignments	Dr. R.J. Rutte	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Recommended: BK 7554 History of Dutch architecture and art	
Course Contents	The Netherlands is the most densely urbanized country in Europe. Its crowded landscape of greater and smaller, older and younger towns was formed in the course of a millennium. What were the roots of this urban landscape, and how did it develop? And how were landscapes and cities represented in the visual arts?	
Study Goals	After seven weeks every student knows the main characteristics and periods of urbanization and landscape transformation in the Netherlands from the tenth to the twenty-first century, and she/he is able to place these developments in their art-historical, economic, political and demographical context. She/he understands that the form of spatial patterns depends on specific historical processes and can answer questions such as: Why do todays Dutch towns and cities look the way they do? And how does the visual representation of landscapes and cities of the past influence the way we perceive them today?	
Education Method	Lectures Excursions	
Literature and Study Materials	R. Rutte & J.E. Abrahamse (eds.), Atlas of the Dutch Urban Landscape. A Millennium of Spatial Development, Bussum/Delft, 2016, p. 156-271, or R. Rutte & J.E. Abrahamse (red.), Atlas van de verstedelijking in Nederland. 1000 jaar ruimtelijke ontwikkeling, Bussum/Delft, 2014, p. 154-269. B. Vannieuwenhuyze & R. Rutte, Medieval urban form in the Low Countries. State of research, comparative perspective and symbolic meaning, in: A. Simms & H.B. Clarke (eds.), Lords and Towns in Medieval Europe. The European Historic Towns Atlas Project, Farnham, 2015, 375-398, or R. Rutte & B. Vannieuwenhuyze, Stadswording in de Lage Landen van de tiende tot de vijftiende eeuw. Een overzicht aan de hand van vijfhonderd jaar ruimtelijke ontwikkeling, Bulletin van de Koninklijke Nederlandse Oudheidkundige Bond 113 (2014) 3, p. 113-131. More specific literature will be provided during the course.	
Assessment	written examination	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter	
Course evaluation	For the course evaluations see: http://kwaliteitszorg.bk.tudelft.nl/	

BK7555	City and Transformation	5
Course Coordinator	Dr. I. Nevzgodin	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	Continuity and change are the keywords for this minor. The transformation of our built environment is of all ages and is currently an urgent design task. In this Research Studio "City and Transformation" you will learn to understand and to develop concepts for the transformation of urban areas and public spaces. The basis for these interventions is a historic analysis of the site and knowledge of architectural history. The challenge is to preserve monumental values, existing qualities and the sense of place while giving way for an intervention that allows for a new function and prepares a site for an up-to-date use. Studies of heritage and other subjects form an important framework. You will immerse yourself in the issue of cultural-historic continuity and urban identity. In this minor the particular focus lies on how we currently handle cultural heritage and necessary urban renewal with reference to changing function requirements. Related to the urban renewal we focus on the use of public areas. To approach these complex tasks, you will be challenged to familiarise yourself with the specific analytical skills and knowledge of city and architectural development of Dutch cities. In addition, you will delve into analysis methods and various theoretic approaches, as well as in cultural, societal and philosophical aspects. You will have the opportunity to gain knowledge in and to practise with aspects such as historical identity, value assessment and spatial intervention with objects and structures of cultural heritage in a historic environment. This minor is intended for all students of TU Delft, Erasmus University and University of Leiden. As both students of architecture and other disciplines participate, the analyses will be developed and presented at a conceptual level, for which it is not necessary to have all the knowledge and skills required for a conventional architectural design. We aim to work in trans-disciplinary teams, where all team members can bring in their creativity, visions and ambitions on an equal base. External academic: art history, architectural history, archaeology, landscape architecture, planning. Level: end second year BSc.	
Study Goals	You will practice the interaction between history and design. At the end of this course you will have enhanced/improved your: knowledge about cultural heritage in the form of urban projects and structures; knowledge about relevant processes in urban history and development; to use professional terminology; knowledge about approaches and work methods concerning appreciation, repurpose and redevelopment of historically valuable objects within an old town; ability to analyse heritage buildings and ensembles with regards to the urban context and its historic development, the functionality, the materials used and the technical development; ability to put historic research and contextual analysis in functional relation with a design task; to apply different tools of spatial analysis and ability to explain your choice of them; ability to put historic research and contextual analysis in functional relation with a design task; ability to justify your point of view in discussions about the valuation of heritage on different scales and about the merging of old and new in case of an intervention.	
Education Method	Lectures, excursions and studio assignments. In the studios, you will work on the research in small groups. For the urban analysis assignments, we will work in a trans-disciplinary manner, using a variety of design skills (architectural design, sketching, mood boards, collage, models, etc.).	
Assessment	The studio assignments are rounded off with the presentation of the analysis reports. The minor is successfully completed if you complete all study parts with grade 6 or higher.	
Special Information	The maximum marking period is 10 work days. The programme of the Minor Heritage%Design I consists of a lecture series and two studio assignments, they account for a total of 15 study points whereby each part is worth 5 study points. BK7551 History of Dutch Cities and Landscapes (lecture series) The development of the Dutch city and the Dutch landscape. BK7555 City and Transformation (studio assignments) The first research studio City and Transformation focusses on the analysis of the city and its buildings. The analysis must ultimately result in a cultural-historic value proposition. This value proposition is used as boundary requirement for a design to be created in the subsequent studios. BK7550 Landscape and Transition (studio assignments) For the design studio Landscape and and Transition, a design must be made of the intervention of a public space, in a historic urban or landscape environment. The contact hours for the studio assignment is 6 hour per week. The contact hours for the lecture series is ca. 4 hour per week. Included self-study the student is supposed to spend (15x28=) 420 hour on the whole course.	
Period of Education	The minor Heritage & Design is offered in the first semester. The different subjects of the program will be divided over the semester.	
Course evaluation	For the course evaluations see: http://kwaliteitszorg.bk.tudelft.nl/	

Year	**2022/2023**
Organization	**Architecture and the Built Environment**
Education	**Minors BK**

BK-MI-212-22

BK7552	Heritage: Theory and Practice	5
Course Coordinator	Dr. I. Nevzgodin	
Instructor	A.C. de Ridder	
Instructor	Dr. I. Nevzgodin	
Responsible for assignments	A.C. de Ridder	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Continuity and change are the keywords for this minor. The transformation of our built environment is of all ages and is currently an urgent design task. In this minor you will learn to develop concepts for the transformation of buildings, urban areas and public spaces. The basis for these interventions is a historic analysis of the site and knowledge of architectural history. The challenge is to preserve monumental values, existing qualities and the sense of place while giving way for an intervention that allows for a new function and prepares a site for an up-to-date use. Studies of heritage and other subjects form an important framework. You will immerse yourself in the issue of cultural-historic continuity and urban identity. In this minor the particular focus lies on how we currently handle cultural heritage and necessary urban renewal with reference to changing function requirements. To approach these complex tasks, you will be challenged to familiarise yourself with the specific design skills and knowledge of city and architectural development of Dutch cities. In addition, you will delve into analysis methods and various theoretic approaches, as well as in cultural, societal and philosophical aspects. You will have the opportunity to gain knowledge in and to practise with aspects such as historical identity, value assessment and spatial intervention with objects and structures of cultural heritage in a historic environment. History is used as inspiration for the design. This minor is intended for all students of TU Delft, Erasmus University and University of Leiden. As both students of architecture and other disciplines participate, the designs will be developed and presented at a conceptual level, for which it is not necessary to have all the knowledge and skills required for a conventional architectural design. We aim to work in trans-disciplinary teams, where all team members can bring in their creativity, visions and ambitions on an equal base. External academic: art history, architectural history, archaeology, landscape architecture, planning. Level: end second year BSc.	
Study Goals	You will learn about the concept of cultural heritage, and the interaction between history and design. At the end of the minor you will have enhanced/improved your: knowledge about different concepts and scale levels of heritage and preservation, from academic research to architectural practice. knowledge about relevant processes in urban history and development. knowledge about approaches and work methods concerning appreciation, re-purpose and redevelopment of historically valuable objects within an old town. understanding of Architectural Heritage and relevant terminology (e.g. Values of Heritage, Intangible Cultural Heritage, Integrity, Reversibility, and Authenticity, etc.). ability to analyse heritage buildings and ensembles with regards to the urban context and its historic development, the functionality, the materials used and the technical development. ability to justify your point of view in discussions about the valuation of heritage on different scales and about the merging of old and new in case of an intervention.	
Education Method	Lectures series and two workshops.	
Assessment	The lecture series are rounded off with an written exam. The group report of the workshops values 10% for the individual final grade. The minor is successfully completed if you complete all study parts with grade 6 or higher.	
Special Information	The maximum marking period is 10 work days. The programme of the Minor Heritage&Design II consists of two lecture series and a design studio assignment, they account for a total of 15 study points whereby each part is worth 5 study points. BK7554 History of Dutch architecture and art (lecture series) In this series of lectures and excursions the stylistic, typological and iconographic developments of Dutch art and architecture from the tenth to the twenty-first century will be studied in close detail. BK7552 Heritage: Theory and Practice (lecture series) Different concepts and scale levels of heritage and preservation, from academic research to architectural practice. BK7553 Architecture and Re-use (studio assignments) For the studio assignment you will work on the transformation of a historic building to which a new function is added. The contact hours for the studio assignments is 6 hours per week. The contact hours for the lecture series is ca. 8 hours a week. Included self-study the student is supposed to spend(15x28=) 420 hours on the whole course.	
Period of Education	The minor Heritage & Design is offered in the first semester. The different subjects of the program will be divided over the semester.	
Course evaluation	For the course evaluations see: http://kwaliteitszorg.bk.tudelft.nl/	

BK7553	Architecture and Re-use	5
Course Coordinator	Dr. I. Nevzgodin	
Course Coordinator	Ir. A.C. de Ridder	
Responsible for assignments	Ir. A.C. de Ridder	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Continuity and change are the keywords for this minor. The transformation of our built environment is of all ages and is currently an urgent design task. In this minor you will learn to develop concepts for the transformation of buildings, urban areas and public spaces. The basis for these interventions is a historic analysis of the site and knowledge of architectural history. The challenge is to preserve monumental values, existing qualities and the sense of place while giving way for an intervention that allows for a new function and prepares a site for an up-to-date use. Studies of heritage and other subjects form an important framework. You will immerse yourself in the issue of cultural-historic continuity and urban identity. In this minor the particular focus lies on how we currently handle cultural heritage and necessary urban renewal with reference to changing function requirements. Related to the urban renewal we also focus on the use of public areas as well as the interior spaces of historic buildings in an old city centre. To approach these complex tasks, you will be challenged to familiarise yourself with the specific design skills and knowledge of city and architectural development of Dutch cities. In addition, you will delve into analysis methods and various theoretic approaches, as well as in cultural, societal and philosophical aspects. You will have the opportunity to gain knowledge in and to practise with aspects such as historical identity, value assessment and spatial intervention with objects and structures of cultural heritage in a historic environment. History is used as inspiration for the design. This minor is intended for all students of TU Delft, Erasmus University and University of Leiden. As both students of architecture and other disciplines participate, the designs will be developed and presented at a conceptual level, for which it is not necessary to have all the knowledge and skills required for a conventional architectural design. We aim to work in trans-disciplinary teams, where all team members can bring in their creativity, visions and ambitions on an equal base. External academic: art history, architectural history, archaeology, landscape architecture, planning. Level: end second year BSc.	
Study Goals	You will practice the interaction between history and design. At the end of the minor you will have enhanced/improved your: knowledge about relevant processes in urban history and development; knowledge about cultural heritage in the form of urban projects and building structures; knowledge about approaches and work methods concerning appreciation, repurpose and redevelopment of historically valuable objects within an old town; ability to analyse complex heritage buildings and ensembles with regards to the urban context and its historic development, the functionality, the materials used and the technical development; understand and deal with complexity of the monument; ability to make architectural and spatial design choices on different scales, based on an analysis and evaluation of current historic information; ability to put historic research and contextual analysis in functional relation with a design task; ability to justify your point of view in discussions about the valuation of heritage on different scales and about the merging of old and new in case of an intervention.	
Education Method	One introduction lecture, the lectures from parallel course BK7552, and studio assignments. In the studios you will work on your own design proposal, using a variety of design skills (architectural design, sketching, mood boards, collage, models, etc.).	
Assessment	The studio design assignments are rounded off with the presentation of design proposals for interventions in an historical building. The minor is successfully completed if you complete all study parts with grade 6 or higher.	
Special Information	The maximum marking period is 10 work days. The programme of the Minor Heritage&Architecture II consists of two lecture series and one design studio assignment, they account for a total of 15 study points whereby each part is worth 5 study points. BK7554 History of Dutch architecture and art (lecture series) In this series of lectures and excursions the stylistic, typological and iconographic developments of Dutch art and architecture from the tenth to the twenty-first century will be studied in close detail. BK7552 Heritage: Theory and Practice (lecture series) Different concepts and scale levels of heritage and preservation, from academic research to architectural practice. BK7553 Architecture and Re-use (studio assignments) For the studio assignment you will work on the transformation of a historic building to which a new function is added. The contact hours for the studio assignments is 6 hours per week. The contact hours for the lecture series is ca. 8 hours a week. Included self-study the student is supposed to spend(15x28=) 420 hours on the whole course.	
Period of Education	The minor Heritage & Design is offered in the first semester. The different subjects of the program will be divided over the semester.	
Course evaluation	For the course evaluations see: http://kwaliteitszorg.bk.tudelft.nl/	

BK7554	History of Dutch architecture and art	5
Course Coordinator	mr.dr. E. Korthals Altes	
Responsible for assignments	mr.dr. E. Korthals Altes	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Recommended: BK7551 History of Dutch Cities and Landscapes	
Course Contents	In this series of lectures and excursions the stylistic, typological and iconographic developments of Dutch architecture and art from the tenth to the twenty-first century will be studied in close detail.	
Study Goals	After seven weeks every student knows the main characteristics and periods of Dutch architecture and art in the Netherlands from the tenth to the twenty-first century, and she/he is able to place these developments in their cultural, economic and political context. She/he understands that form and function of buildings and art works depend on specific historical processes and can answer questions such as: Why does the architecture of Dutch towns and cities look the way it does today? And how does the visual representation of the architecture of the past influence the way we perceive the built environment today?	
Education Method	Lectures and excursions	
Literature and Study Materials	R.H. Fuchs, Dutch Painting, 1978 (or later editions). Additional literature will be provided during the course. More specific literature will be provided during the course.	
Books	R.H. Fuchs, Dutch Painting, 1978 (or later editions).	
Prerequisites	Advanced Bachelor (HBO 3 years WO 2 years).	
Assessment	Written examination	
Exam Hours	2	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 2	
Course evaluation	For the course evaluations see: http://kwaliteitszorg.bk.tudelft.nl/	

Year	**2022/2023**
Organization	**Architecture and the Built Environment**
Education	**Minors BK**

BK-MI-2016-22

BK7215	Basic techniques for sustainable urban design	5
Course Coordinator	Ir. M.G.F. Overschie	
Responsible for assignments	Ir. M.G.F. Overschie	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	In a series of interactive lectures and workshops, present and future urban tasks will be presented and discussed, and urban design principles for sustainable urban plans will be practiced. Innovative realized plans will be introduced as examples. Sustainable urbanism themes such as climate change, the water challenge, heat stress, biodiversity, ecology, urban agriculture, air quality, energy, mobility, urban transport, waste and material flows, social and economic value, and process will be discussed. By field trips future proof sustainable plans will be illustrated. The edX MOOC Sustainable Urban Development will provide basic knowledge on global challenges and sustainable urbanism themes. In the urbanism professions, visualizations play an important role. They serve to communicate research results and planning proposals. But they are also used to develop ideas, in brainstorming and design. In this course, the use of visual means at each stage of the research and design process will be discussed. Students will work on visualisation assignments, such as: sketching to develop and share our ideas; mapping to understand the conditions, the history and to orient ourselves; and info graphs to explain an idea or to present statistics.	
Study Goals	After finishing this course, students: * Have the ability to recognize sustainable plans and sustainable measures (people, planet and profit) in urban plans. * Have knowledge of sustainable urbanism themes, e.g climate change effects and the spatial implications of adapting to climate change in sustainable urban plans. * Have basic skills to understand and use visual instruments, e.g. sketching, mapping and using icons.	
Education Method	Series of lectures, workshops and field trips (in and around Delft by bike), combined with self-study (edX MOOC Sustainable Urban Development).	
Assessment	* Take home exam on the content of the lectures, workshops, videos, and literature displayed on Brightspace. * Complete self-paced edX MOOC Sustainable Urban Development.	
Enrolment / Application	This course is part of the minor Sustainable Urbanism.	
Special Information	The maximum marking period of the take home exam is 15 work days.	
Period of Education	2nd quarter (re-take of the take home exam in 3th quarter)	
Leerstoel	Environmental Technology & Design	
Maximum number of participants	30	

BK7225	City and Public Space: Sustainable Urban Design	10
Course Coordinator	Ir. L.P.J. van den Burg	
Responsible for assignments	Ir. L.P.J. van den Burg	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	This part of the minor "Sustainable Urbanism: the Green Blue City" is integrated in the framework of the regular Bsc-3 design module ON3 in the environmental design variant. Course overview The programme consists of two courses of a total of 15 ECs. City and Public Space: Sustainable Urban Design (studio) (BK7225): 10 ECs Basic Techniques for Sustainable Urban Design: 5 ECs The task of the urban planner is to introduce and design sustainable urban interventions, creating future proof neighbourhoods within the framework of a green blue city. In the Sustainable Urban Design studio City and Public Space of this minor a sustainable urban design is the main subject. In this studio the minor students will develop design proposals, in interdisciplinary teams. Focus in the studio is on concrete sustainable design exercises. The course Basic Techniques for Sustainable Urban Design will make the minor students equipped and prepared to perform well in the design studio. They will apply theory and use their knowledge background in the group design process. In this minor you will learn: to recognize inner-city problems caused by climate change and social aspects; to recognize sustainable urban plans and sustainable measures in urban plans; use knowledge of climate change effects and the spatial implications of adapting to climate change in sustainable urban plans; to develop, integrate and evaluate spatial interventions for a future proof urban plan; to collaborate with students from other disciplines in order to improve own (basic) urban design skills, research skills and writing skills, and help improve these skills of others; to argue comprehensively for choices made in the design process, considering spatial, social and procedural circumstances; to give relevant feedback on content and presentation to peers, and accept and react on received feedback. For whom? The minor is intended for students from urban planning, regional planning, civil engineering, landscape architecture, technology policy and management, industrial design engineering, social sciences, and other studies related to the topic of climate change and the city. The minor is closed for architecture students from Delft University of Technology, however open to architecture exchange students. You must have interest and/or affection for design as may be expected from your respective expertise and bachelor programme. Furthermore, you have an interest in social and sustainable aspects and you are academically curious. This minor has an interdisciplinary character and thus will assemble students in groups to create a balanced variety of faculty backgrounds. For students from outside Delft University of Technology, Leiden University or Erasmus University a specific selection procedure applies. After applying as described on the minor website you will receive an email within three weeks in which you will be asked to fill in a questionnaire. After receiving your questionnaire, you will be informed whether you will be officially enrolled or not.	
Study Goals	General; the student - is able to make a design for a wide range of assignments, taking into account the relevant requirements and preconditions; - is able to make a design based on self-formulated principles, using a wide range of research techniques and is able to critically reflect on the quality of the design; - understands a wide range of basic technical requirements, sees them as an intrinsic part of designing and applies them as such. Specific; the student - is able to dissect the urban landscape into networks, allotment patterns, types of buildings and public spaces and is able to integrate these in a sustainable design, on different scales; - is able to control multiple scales simultaneously and in conjunction in the design. The student shows an understanding of the relationship between the main urban networks, allotment patterns and block typology. - can make urban design analyses based on design questions. The student can design in variants and relate these to the results of the analyses. The student is able to make calculations with regard to densities and take the results into account in the design. - can document the design process in urban design impressions, plan maps, profiles and models.	
Education Method	design studio	
Assessment	2 intermediate presentations in week, final presentation in week 9. One final grade is given on the basis of the quality of the final portfolio, deadline in week 10. If the course is graded as insufficient, the whole course has to be repeated at a later time or in the two-week summer school. The latter option is open only for students with a grade 5 of 5,5	
Period of Education	2	

Year	**2022/2023**
Organization	**Architecture and the Built Environment**
Education	**Minors BK**

BK-MI-231-22 (Re)Imagining Port Cities: Understanding Space, Society and Culture

BK7940	(Re)imagining Port Cities: Understanding Space, Society and Culture Q1	15
Course Coordinator	Ir. E.J.G.C. van Dooren	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Designing is imagining the future. In a time of climate change and multiple transitions, a designerly attitude is needed more than ever. At the same time, understanding past practices and socio-cultural conditions is an essential basis for new interventions. In the first quarter of the minor, the port city is studied from cultural, social and historical perspectives. Various topics are covered in lectures. Research and design studies are done for each of the topics. In order to learn to think like a designer, the basic skills of the design process are made explicit as much as possible.	
Study Goals	In this course, you will work on a series of short design and research studies on different themes and learn to: study (analyze and reflect) the historical, social and cultural context in port-city areas in relation to the present and the future, experiment (explore and test) with possible alternative future ideas, supported by the studies, take position as designer regarding the future of port cities.	
Education Method	lectures and tutorials in the (design) studio	
Assessment	Assessment will be based on the research and design studies, a combination of text and means to present a design, such as images, drawings and sketches.	
Period of Education	quarter 1	

BK7941	(Re)imagining Port Cities: Understanding Space, Society and Culture Q2	15
Course Coordinator	Ir. E.J.G.C. van Dooren	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Designing is imagining the future. In a time of climate change and multiple transitions, a designerly attitude is needed more than ever. At the same time, understanding past practices and socio-cultural conditions is an essential basis for new interventions. In the second quarter, each student chooses a theme as perspective to (re)imagine the future of a port city. Depending on the theme, it is decided which information items are important to arrive at a design. For these items, some research and design studies are done. Examples of themes are - green city, - rising water level, - social development, - personal choice (in discussion with teacher) To learn to think like a designer, the basic skills of the design process are made explicit as much as possible.	
Study Goals	In this course, you will work on a design project: on research studies and design interventions on different scales regarding one theme, and learn to: study (analyze and reflect) the historical, social and cultural context in port-city areas in relation to the present and the future, experiment (explore and test) with possible alternative future ideas, supported by the studies, take position as designer regarding the future of port cities by choosing a theme or quality, elaborate the theme in a design at different scale levels, such as public space, street furniture and/or an urban vision.	
Education Method	lectures and tutorials in the (design) studio	
Assessment	Assessment will be based on the research and design studies, a combination of text and means to present a design, such as images, drawings and sketches.	
Period of Education	quarter 2	

Year	**2022/2023**
Organization	**Architecture and the Built Environment**
Education	**Minors BK**

BK-MI-235-22 Spatial Computing for Digital Twinning

BK7944	Spatial Computing for Digital Twinning	15
Course Coordinator	Dr. A. Rafiee	
Instructor	Dr.ir. B. van Loenen	
Instructor	Dr. R.C. Lindenbergh	
Instructor	Mr.dr. H.D. Ploeger	
Instructor	Ir. J.A. van Hooft	
Instructor	Dr. A. Rafiee	
Responsible for assignments	Dr. A. Rafiee	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	This minor part contains three courses, each one with 5 ECTS: Course: 3D Data Acquisition & Analysis This course provides students with an overview and hands-on experience with 3D spatial data acquisition techniques and methodology for information extraction from 3D data, linked to urban applications. The course will consist of a combination of lectures and peer-to-peer presentations by students. Students work in groups of three on an integrating assignment around a relevant topic about acquisition, information extraction and application. The individual level of the students is assessed by means of a short written exam. Course: Open urban data governance This course aims to provide a general insight into the relevant legislation and regulations as well as organizational aspects involved in the collection, processing, provision and use of individual datasets in the built environment. In small groups, the students will answer a specific social sustainability related question in the context of the built environment with open urban (geo) data. Course: Spatial Decision Support Systems The course provides a basis for spatial analysis in a large area. Students learn how to model and visualize the mutual impact of the environment and their design. Students also learn about solving complex spatial decision problems at a geographic scale, for which they apply various simulation and optimization approaches, among other things. The assignments are related to more sustainable development of the living environment, such as location allocation/planning of wind turbines and solar panels. The course will consist of a combination of lectures and hands-on labs. Students work in groups of 2-3 on an assignment around a complex spatial decision problem. The individual level of the students is assessed by means of the exam.	
Study Goals	Course: 3D Data Acquisition & Analysis After completing the 3D Data Collection and Analysis course, students will be able to: - Explain the principles and properties of 3D point cloud acquisition techniques, including in particular photogrammetry and laser scanning - Draw conclusions about the accuracy and precision obtained with different measurement setups and processing techniques - Extract information from point clouds in a suitable computing environment - Discuss the advantages and disadvantages of different techniques for extracting information from point clouds Course: Open urban data governance After taking this course, students are expected to be able to: - Relevant national and international legislation and policy (data processing, legislation regarding the accessibility of data (Open Government Act, Land Registry Act, INSPIRE Directive) and organizational concepts (open data, FAIR data, master data, data management) regarding the collection, processing, provision and use of individual datasets, - Provide insight into data user needs and report on them - Recognize possible opportunities and limitations in the use of the data from a legal and organizational perspective. - To propose solutions for a specific case with regard to the design of the city in order to design a framework for optimal use of the data. Course: Spatial Decision Support Systems After the course the student is able to: - Understand the complexity of spatial problems and the role of Spatial Decision Support Systems (SDSS) - Explain the concept, components and underlying technologies of SDSS - Apply Multicriteria Decision Analysis (MCDA) for spatial decisions - Use Multi-criteria Spatial Decision Support System (MC-SDSS) when making decisions about the (mutual) impact of a design and its environment - Compare design or decision alternatives according to the design environment effects and evaluation criteria	
Education Method	Course: 3D Data Acquisition & Analysis - Lectures/labs including peer presentations: 40 hours - 40 hours assignments; - 60 hours self-study: The lectures and small assignments focus on knowledge and insight and require active student participation The group project (i.e. big assignment) focuses on learning by doing and inquiry based knowledge acquisition with 2-3 students per group Course: Open urban data governance Lectures (14 hours)/ group work (86 hours)/self study (40 hours) Course: Spatial Decision Support Systems Lectures/labs 32 hours; 26 hours Assignments; 82 hours self-study. The lectures focus on Knowledge and Insight. The group projects focus on Applying the Knowledge and Insight. For the group projects, 2-3 students per group will work together.	
Assessment	Course: 3D Data Acquisition & Analysis Big assignments and a small written exam. The result of the big assignment determines 60% of the final score and the small written exam 40%. Course: Open urban data governance Groups of two students write a paper about their case study and there is an open book exam.	

	Course: Spatial Decision Support Systems Written exam and group assignments. The result of the written exam determines 50% of the final score and the group assignments 50%.
Period of Education	Quarter 1

Year	**2022/2023**
Organization	**Architecture and the Built Environment**
Education	**Minors BK**

BK-MI-236-22 Timeless Typologies - "Remodelling Architecture"

BK7942	Culture & Understanding	5
Course Coordinator	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	5 hours per week	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The minor Timeless Typologies will focus on studying and modelling historical architectural precedents, covering a known oeuvre or style. The chosen theme will be represented by a series of case study projects, regarded as well-respected in the architectural field. The main goal is for students to understand which historical, spatial and technical values of the architectural precedents can be related to present architectural practice. The courses Culture & Understanding (History) and Design & Remodelling (Form Studies) are both integrated in this minor.	
Study Goals	At the end of the minor you: have acquired knowledge, insights, and skills in the domains of architectural research, design, analysis, and presentation; have developed abilities in the targeted analysis of design precedents and the effective representation and communication of findings, using various representation techniques, with a focus on physical models regarding Design & Remodelling have the ability to work individually and in a team, carrying out research, documenting results, visualizing proposals and bringing these across to a professional audience.	
Education Method	Lectures, workshops, feedback sessions, physical modelling : all in design studio format.	
Assessment	For Culture & Understanding: Essay, based on first impressions and literature research. Research report. Written dossier, relating the project to essay in a reflective matter.	
Period of Education	Quarter	

BK7943	Design & Remodelling	10
Course Coordinator	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The minor Timeless Typologies will focus on studying and modelling historical architectural precedents, covering a known oeuvre or style. The chosen theme will be represented by a series of case study projects, regarded as well-respected in the architectural field. The main goal is for students to understand which historical, spatial and technical values of the architectural precedents can be related to present architectural practice. The courses Culture & Understanding (History) and Design & Remodelling (Form Studies) are both integrated in this minor.	
Study Goals	At the end of the minor you: have acquired knowledge, insights, and skills in the domains of architectural research, design, analysis, and presentation; have developed abilities in the targeted analysis of design precedents and the effective representation and communication of findings, using various representation techniques, with a focus on physical models regarding Design & Remodelling have the ability to work individually and in a team, carrying out research, documenting results, visualizing proposals and bringing these across to a professional audience.	
Education Method	Lectures, workshops, feedback sessions, physical modelling : all in design studio format.	
Assessment	For Design & Remodelling Analysis in sketch models, drawings and written group manifest. Presentation: Models, posters, oral in exhibition setting.	
Period of Education	Quarter	

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